THE LOCAL RINGS OF A SCHEME AND OF ITS BASE CHANGE TO THE COMPLETION HAVE ISOMORPHIC COMPLETIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to the affine case and the comparison of the quotients by the powers of the prime ideals via the identified tensor product quotients.

Editorial note. The label lemma-iso-completions names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-iso-completions –start 2965 –end 2994 –local-id 28113 returned RESULT=OK (L28113 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Base change of a tensor product along the completion modulo powers of the maximal ideal. Algebra, the lemma identifying the tensor product of the completion with the base modulo the powers of the maximal ideal. Identifies the quotients of the tensor product ring appearing in the computation of the nth powers of the prime ideals.

Source reference. Original lemma 05GG: sources/stacks-chow/tags.txt, source lines 7060–7060. Prerequisite pointer invoked at proof line(s) 2982; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a local ring with finitely generated maximal ideal $\mathfrak{m}$. Let X be a scheme over A. Let $Y = X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A^\wedge)$ where $A^\wedge$ is the $\mathfrak{m}$-adic completion of A. For a point $q \in Y$ with image $p \in X$ lying over the closed point of $\mathrm{Spec}(A)$ the local ring map $\mathcal{O}_{X,p} \to \mathcal{O}_{Y,q}$ induces an isomorphism on completions.*

Proof. We may assume X is affine. Then we may write $X = \mathrm{Spec}(B)$. Let $\mathfrak{q} \subset B' = B \otimes_A A^\wedge$ be the prime corresponding to q and let $\mathfrak{p} \subset B$ be the prime ideal corresponding to p. By Algebra, Lemma 05GG we have

$$B'/(\mathfrak{m}^\wedge)^n B' = A^\wedge/(\mathfrak{m}^\wedge)^n \otimes_A B = A/\mathfrak{m}^n \otimes_A B = B/\mathfrak{m}^n B$$

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for all n. Since $\mathfrak{m}B \subset \mathfrak{p}$ and $\mathfrak{m}^\wedge B' \subset \mathfrak{q}$ we see that $B/\mathfrak{p}^n$ and $B'/\mathfrak{q}^n$ are both quotients of the ring displayed above by the nth power of the same prime ideal. The lemma follows. $\square$